In [2]:
```python
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import Sequential
from keras.layers import Dense,Conv2D,MaxPooling2D,Flatten,Dropout,BatchNo
```

```
WARNING:tensorflow:From C:\Users\denni\AppData\Local\Programs\Python\Pyth
on311\Lib\site-packages\keras\src\losses.py:2976: The name tf.losses.spar
se_softmax_cross_entropy is deprecated. Please use tf.compat.v1.losses.sp
arse_softmax_cross_entropy instead.
```

In [3]:
```python
train_ds = keras.utils.image_dataset_from_directory(
    directory = r'D:\SASTRA\2023 to 2024 AY\ODD SEM\Deep Learning\DL EXPER
    labels = 'inferred',
    label_mode = 'int',
    batch_size = 32,
    image_size = (256,256))

validation_ds = keras.utils.image_dataset_from_directory(
    directory = r'D:\SASTRA\2023 to 2024 AY\ODD SEM\Deep Learning\DL EXPER
    labels = 'inferred',
    label_mode = 'int',
    batch_size = 32,
    image_size = (256,256))
```

```
Found 8005 files belonging to 2 classes.
Found 2023 files belonging to 2 classes.
```

In [4]:
```python
#Normalize
def process(image,label):
    image = tf.cast(image/255., tf.float32)
    return image,label

train_ds = train_ds.map(process)
validation_ds = validation_ds.map(process)
```

In [5]:

```
#create CNN model

model = Sequential()

model.add(Conv2D(32,kernel_size=(3,3),padding='valid',activation='relu',in
model.add(BatchNormalization())
model.add(MaxPooling2D(pool_size=(2,2),strides=2,padding='valid'))

model.add(Conv2D(64,kernel_size=(3,3),padding='valid',activation='relu'))
model.add(BatchNormalization())
model.add(MaxPooling2D(pool_size=(2,2),strides=2,padding='valid'))

model.add(Conv2D(128,kernel_size=(3,3),padding='valid',activation='relu'))
model.add(BatchNormalization())
model.add(MaxPooling2D(pool_size=(2,2),strides=2,padding='valid'))

model.add(Flatten())

model.add(Dense(128,activation='relu'))
model.add(Dropout(0.1))
model.add(Dense(64,activation='relu'))
model.add(Dropout(0.1))
model.add(Dense(1,activation='sigmoid'))
```

```
WARNING:tensorflow:From C:\Users\denni\AppData\Local\Programs\Python\Pyth
on311\Lib\site-packages\keras\src\backend.py:873: The name tf.get_default
_graph is deprecated. Please use tf.compat.v1.get_default_graph instead.

WARNING:tensorflow:From C:\Users\denni\AppData\Local\Programs\Python\Pyth
on311\Lib\site-packages\keras\src\layers\normalization\batch_normalizatio
n.py:979: The name tf.nn.fused_batch_norm is deprecated. Please use tf.co
mpat.v1.nn.fused_batch_norm instead.
```

In [6]:
```
model.summary()
```

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 conv2d (Conv2D)             (None, 254, 254, 32)      896

 batch_normalization (Batch  (None, 254, 254, 32)      128
 Normalization)

 max_pooling2d (MaxPooling2  (None, 127, 127, 32)      0
 D)

 conv2d_1 (Conv2D)           (None, 125, 125, 64)      18496

 batch_normalization_1 (Bat  (None, 125, 125, 64)      256
 chNormalization)

 max_pooling2d_1 (MaxPoolin  (None, 62, 62, 64)        0
 g2D)

 conv2d_2 (Conv2D)           (None, 60, 60, 128)       73856

 batch_normalization_2 (Bat  (None, 60, 60, 128)       512
 chNormalization)

 max_pooling2d_2 (MaxPoolin  (None, 30, 30, 128)       0
 g2D)

 flatten (Flatten)           (None, 115200)            0

 dense (Dense)               (None, 128)               14745728

 dropout (Dropout)           (None, 128)               0

 dense_1 (Dense)             (None, 64)                8256

 dropout_1 (Dropout)         (None, 64)                0

 dense_2 (Dense)             (None, 1)                 65

=================================================================
Total params: 14848193 (56.64 MB)
Trainable params: 14847745 (56.64 MB)
Non-trainable params: 448 (1.75 KB)
_________________________________________________________________
```

In [7]:
```
model.compile(optimizer='adam',loss='binary_crossentropy',metrics=['accura
```

```
WARNING:tensorflow:From C:\Users\denni\AppData\Local\Programs\Python\Pyth
on311\Lib\site-packages\keras\src\optimizers\__init__.py:309: The name t
f.train.Optimizer is deprecated. Please use tf.compat.v1.train.Optimizer
instead.
```

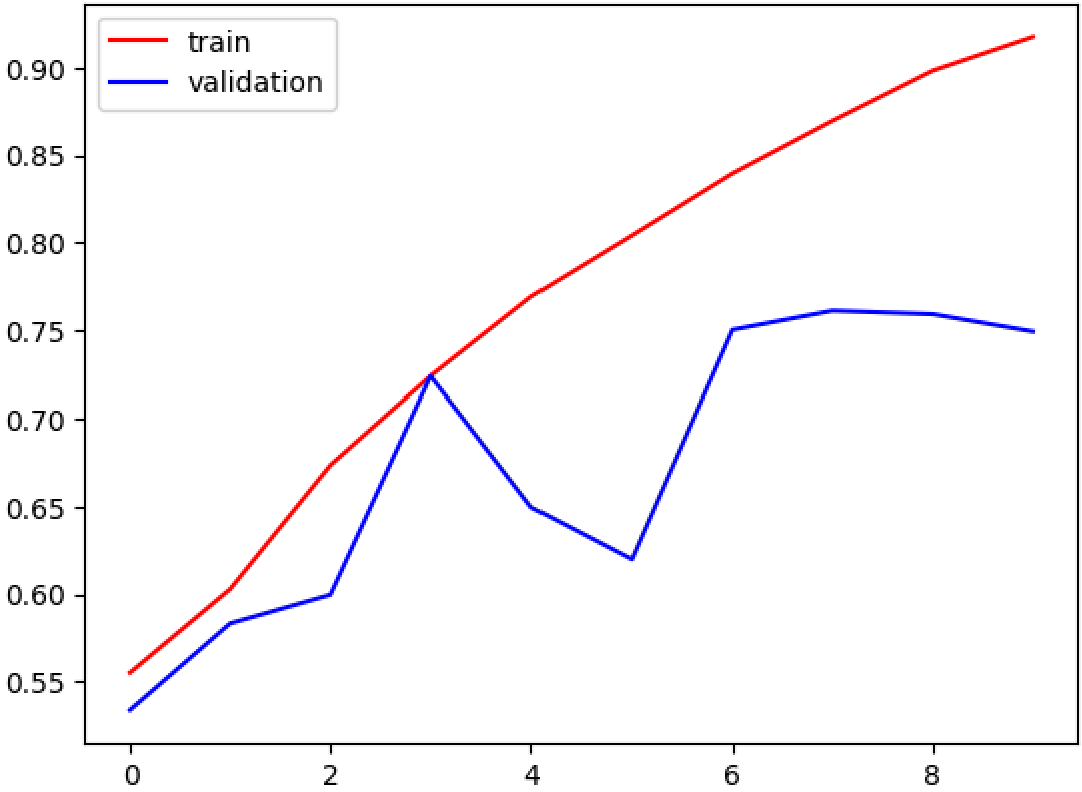
In [ ]:

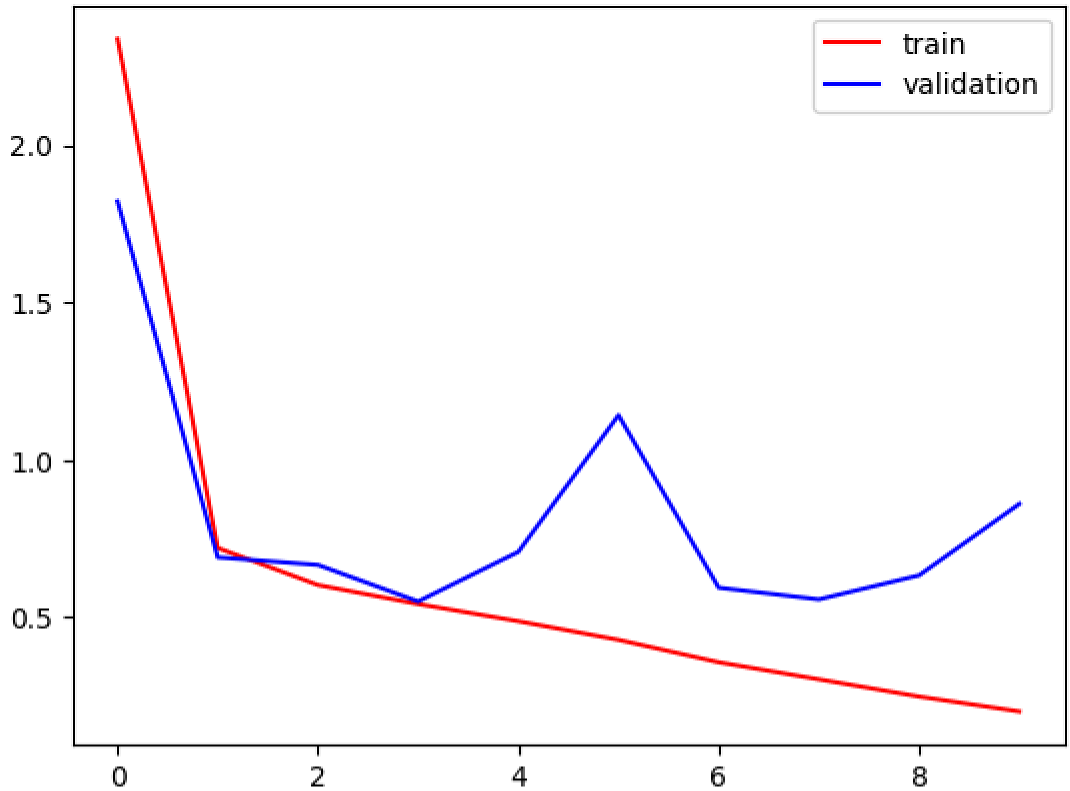
```
history = model.fit(train_ds,epochs=10,validation_data=validation_ds)
```

```
Epoch 1/10
 24/251 [=>...........................] - ETA: 6:45 - loss: 3.7433 - acc
uracy: 0.6133
```

In [8]:

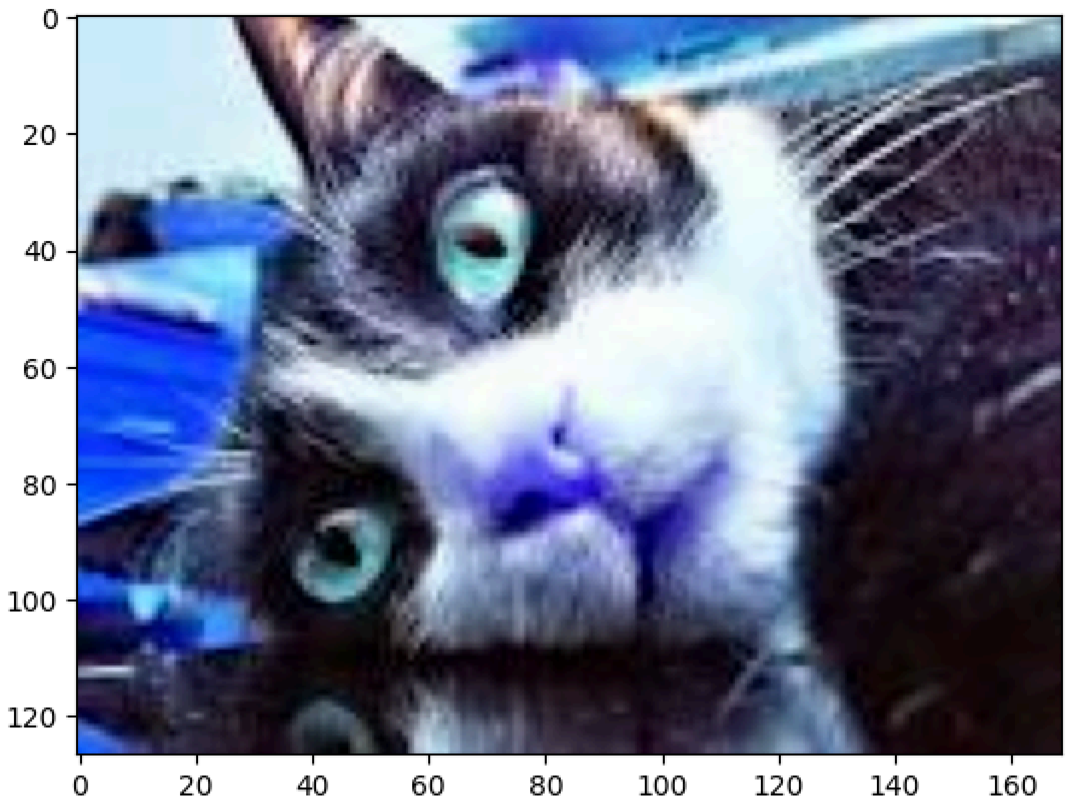
```
import matplotlib.pyplot as plt

plt.plot(history.history['accuracy'],color='red',label='train')
plt.plot(history.history['val_accuracy'],color='blue',label='validation')
plt.legend()
plt.show()

plt.plot(history.history['loss'],color='red',label='train')
plt.plot(history.history['val_loss'],color='blue',label='validation')
plt.legend()
plt.show()
```





In [15]:
```
import cv2
test_img = cv2.imread('C:/Users/denni/OneDrive/Desktop/Image classificatio
plt.imshow(test_img)
```

Out[15]: <matplotlib.image.AxesImage at 0x1d6b848be90>



In [16]:
```
test_img.shape
```
Out[16]:
```
(127, 169, 3)
```
In [17]:
```
test_img = cv2.resize(test_img,(256,256))
```
In [18]:
```
test_input = test_img.reshape(1,256,256,3)
```
In [19]:
```
model.predict(test_input)
```
```
1/1 [==============================] - 0s 37ms/step
```
Out[19]:
```
array([[0.]], dtype=float32)
```
In [ ]:
```

```